

# Docker Containers Assignment Submission

## Commands Used to Build and Run the Stack

### Main Commands

```
# Go to the project directory  
cd path/to/project
```

```
# Build images and start containers  
make all
```

```
# Shut down the containers  
make down
```

These commands use the Makefile to build the Docker setup, run both services, and stop them when finished.

---

### Alternative Commands (Without Makefile)

```
# Start services directly with Docker Compose  
docker compose up --build
```

```
# Stop and clean up containers and volumes  
docker compose down -v
```

These commands perform the same tasks manually using Docker Compose.

---

### Checking Results

```
# Display generated JSON output  
cat out/summary.json
```

```
# Show running container status  
docker compose ps
```

```
# View logs if troubleshooting is needed  
docker compose logs app
```

These commands help confirm the application ran successfully and produced the expected output.

## Summary Output from Terminal (stdout)

Example output printed when the Python container runs:

```
{
  "total_trips": 6,
  "avg_fare_by_city": [
    {"city": "Charlotte", "avg_fare": 16.25},
    {"city": "New York", "avg_fare": 19.0},
    {"city": "San Francisco", "avg_fare": 20.25}
  ],
  "top_by_minutes": [
    {"city": "San Francisco", "minutes": 28, "fare": 29.30},
    {"city": "New York", "minutes": 26, "fare": 27.10},
    {"city": "Charlotte", "minutes": 21, "fare": 20.00},
    {"city": "Charlotte", "minutes": 12, "fare": 12.50},
    {"city": "San Francisco", "minutes": 11, "fare": 11.20},
    {"city": "New York", "minutes": 9, "fare": 10.90}
  ]
}
```

## Contents of out/summary.json

The same JSON summary is written to:  
out/summary.json

File contents:

```
{
  "total_trips": 6,
  "avg_fare_by_city": [
    {"city": "Charlotte", "avg_fare": 16.25},
    {"city": "New York", "avg_fare": 19.0},
    {"city": "San Francisco", "avg_fare": 20.25}
  ],
  "top_by_minutes": [
    {"city": "San Francisco", "minutes": 28, "fare": 29.30},
    {"city": "New York", "minutes": 26, "fare": 27.10},
    {"city": "Charlotte", "minutes": 21, "fare": 20.00},
    {"city": "Charlotte", "minutes": 12, "fare": 12.50},
    {"city": "San Francisco", "minutes": 11, "fare": 11.20},
    {"city": "New York", "minutes": 9, "fare": 10.90}
  ]
}
```

## **Reflection**

This assignment helped me understand how Docker containers work together in a multi-container environment using Docker Compose. I learned how to run a PostgreSQL database in one container and connect to it from a Python application using environment variables and container networking. I also gained experience with database initialization scripts, volume mounts, and generating reproducible outputs. If I were to improve this project, I would add more complex analytics queries, better error handling, and possibly a small interface to visualize the results.